

Unrolled Iterative Reconstruction with Calibrated Uncertainty for Low-Dose CT: A Reference Method for the PWM Benchmark

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Abstract

We present an open-source reference reconstruction method for low-dose CT designed not to maximize peak signal-to-noise ratio (PSNR) but to anchor a permanent, reproducibility-first benchmark with three properties seldom co-located in published methods: **calibrated per-pixel uncertainty**, **cross-vendor generalization**, and **permissive licensing**. The method is an unrolled iterative reconstruction with 10 data-consistency steps interleaved with a learned U-Net denoiser whose weights are shared across iterations. Uncertainty is quantified via a five-model deep ensemble. The forward model is the open `ct_radon` implementation maintained as part of the PWM modality engine, ensuring that any user replicating the architecture inherits a validated Radon transform without re-implementation. We evaluate on the multi-vendor PWM-LDCT benchmark across four dose levels (10%, 25%, 50%, 100%) and report (i) reconstruction quality (PSNR/SSIM/LPIPS), (ii) downstream task performance (lung-nodule detection AUC), (iii) cross-vendor generalization (train-on-Siemens, test-on-GE), (iv) uncertainty calibration (per-pixel error correlation with predicted variance), and (v) signal-equivalence credentials per the framework of [to be confirmed \(2027b\)](#). The reference method is released under Apache 2.0 with a content-addressed RunBundle on IPFS and a verified L4 cert on the PWM blockchain registry. Our explicit goal is *not* state-of-the-art — top-25% with these five properties is more useful as a community baseline than top-1% without them. The method is designed for follow-on improvement: each component (denoiser, forward model, iteration schedule, uncertainty quantification) can be independently swapped without retraining the entire pipeline, inviting future methods to compete on individual axes rather than re-derive the full stack.

Keywords: Low-dose CT Unrolled iterative reconstruction Uncertainty quantification Cross-vendor generalization Reproducible benchmark.

1 Introduction

Low-dose CT reconstruction has accumulated several years of method papers reporting PSNR improvements of fractions of a decibel over the previous best ([Chen et al., 2017](#); [Wang et al., 2020](#); [to be confirmed at submission, 2024](#)). Yet adoption of any single method as the field’s reference baseline has not occurred. We argue the obstacle is structural: published methods optimize for performance on the dataset used during development (typically AAPM-2016 ([AAPM, 2016](#)), captured on Siemens hardware) but rarely report (i) cross-vendor generalization, (ii) per-pixel uncertainty calibration, (iii) downstream task performance rather than pixel-wise metrics, or (iv) open-source code with permissive licensing. A user picking a method to deploy on a different vendor’s data, or to provide diagnostic confidence on individual cases, has no clear choice.

This paper presents a reference reconstruction method explicitly designed for these four properties, deliberately accepting that PSNR will not be state-of-the-art. The method is positioned as the seed entry of a permanent benchmark leaderboard ([to be confirmed, 2028](#)): a credible top-25% performer that the community can improve upon, on any of four independent axes.

Architecture rationale. We use unrolled iterative reconstruction (Adler and Öktem, 2018; Mardani et al., 2018) rather than purely end-to-end neural reconstruction for three reasons.

1. **Physical prior.** The Radon forward model is exact (modulo discretization). Embedding it in the iteration loop guarantees data consistency between the reconstruction and the measured sinogram, a property that pure end-to-end networks must learn from data.
2. **Decomposable improvability.** A user wishing to improve our reference method has four independent levers — swap the denoiser, swap the forward-model implementation, learn iteration step sizes, replace deep ensembles with a different UQ technique — each of which yields a publishable contribution that strictly improves on this baseline without invalidating its overall architecture. Pure transformer or pure diffusion baselines do not offer this decomposability (to be confirmed at submission, 2024).
3. **Interpretability under regulatory review.** Iterative methods with a clear data-consistency loop have a precedent in regulated clinical reconstruction (vendor iterative reconstruction algorithms have FDA clearance). An open-source reference method that follows the same architectural class is more transferable to regulated deployment paths than a pure neural approach.

Why we name our goal “top-25%”. A reference method should be *credible, improvable, and permissively licensed*. Chasing the last 0.3 dB on a single dataset frequently sacrifices one or more of these properties: methods become brittle, narrowly tuned, or encumbered by patent claims. We adopt the explicit design goal of top-25% on the leaderboard (to be confirmed, 2028) with all four desired properties present; this is a more useful seed for the community than a state-of-the-art single-vendor method without them.

Contributions.

1. An open-source unrolled iterative reconstruction architecture with weight-shared U-Net denoiser, TV warm start, and five-model deep ensemble uncertainty quantification, sized to fit in < 24 GB GPU memory (Section 2).
2. Cross-vendor generalization evaluation: train on Siemens scans from the PWM-LDCT benchmark (to be confirmed, 2027a), test on GE and Canon (Section 3).
3. Per-pixel uncertainty calibration: Spearman correlation between predicted per-pixel variance and observed per-pixel reconstruction error (Section 4).
4. Five-tuple signal-equivalence credentials per the framework of to be confirmed (2027b) for the canonical lung-nodule detection task at 25% and 10% dose (Section 4.1).
5. Release under Apache 2.0 with a reproducible Docker RunBundle published to IPFS and a corresponding L4 cert on the PWM blockchain registry.

2 Method

2.1 Forward model

Let $\mathcal{R} : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$ denote the Radon forward operator and $\mathcal{R}^\dagger$ its filtered back-projection adjoint. Both are implemented in `pwm_core.contrib.modalities.ct_radon` (PWM Protocol, 2026), the validated forward-model implementation used across PWM’s 172-modality engine. We do not re-implement; using the upstream implementation guarantees that bug-fixes propagate uniformly and that any user of our reference method inherits a Radon transform validated against the full PWM modality test suite.

Algorithm 1 Unrolled iterative reconstruction

```
1:  $x^{(0)} \leftarrow \text{TV-FBP}(y)$  ▷ TV-regularized warm start
2: for  $k = 0, 1, \dots, K-1$  do
3:    $x^{(k+1/2)} \leftarrow x^{(k)} - \tau \cdot \mathcal{R}^\dagger(\mathcal{R}(x^{(k)}) - y)$  ▷ Data consistency
4:    $x^{(k+1)} \leftarrow f_\theta(x^{(k+1/2)})$  ▷ Learned denoising
5: end for
6: return  $\hat{x} \leftarrow x^{(K)}$ 
```

2.2 Unrolled iteration

Given a low-dose sinogram $y \in \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$, we estimate the reconstruction $\hat{x} \in \mathbb{R}^{H \times W}$ via K iterations of a data-consistency step interleaved with a learned denoiser f_θ :

We use $K = 10$ iterations; ablation results for $K \in \{5, 10, 15, 20\}$ are in Section 4 Table 1. The denoiser f_θ is a U-Net with ≈ 4 M parameters whose weights are shared across iterations; this both reduces memory and acts as an implicit regularizer (the same f_θ must denoise at all iteration depths, preventing iteration-specific overfitting).

[TODO: TABLE: Iteration-count ablation: PSNR / SSIM on PWM-LDCT test split for $K \in \{5, 10, 15, 20\}$]

Table 1: Iteration-count ablation: PSNR / SSIM on PWM-LDCT test split for $K \in \{5, 10, 15, 20\}$ (placeholder; populate when data lands).

The step size τ is a learned scalar, initialized at $0.5/\|\mathcal{R}\|^2$ (the conventional Landweber-stable value). End-to-end optimization with mean absolute error loss on Hounsfield-unit values:

$$\mathcal{L}(\theta, \tau) = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i(\theta, \tau; y_i) - x_i^{\text{ref}}\|_1 \quad (1)$$

where x_i^{ref} is the patient’s full-dose reconstruction (the reference image).

2.3 Uncertainty quantification

We use a five-model deep ensemble (Lakshminarayanan et al., 2017): five independently-initialized and independently-trained instances of the same architecture, differing only in random seed for weight initialization and data ordering. At inference time, given a low-dose sinogram y , each ensemble member produces a reconstruction $\hat{x}_m(y)$ for $m = 1, \dots, 5$. The point estimate is the mean and the per-pixel uncertainty is the standard deviation:

$$\hat{x}(y) = \frac{1}{5} \sum_{m=1}^5 \hat{x}_m(y), \quad \sigma(y)_{ij} = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{x}_m(y)_{ij} - \hat{x}(y)_{ij})^2}. \quad (2)$$

We choose deep ensembles over Monte-Carlo dropout because (i) ensembles consistently outperform dropout on calibration in regression benchmarks (Ovadia et al., 2019), (ii) inference cost is bounded ($5 \times$ a single model rather than T -stochastic), and (iii) the inference cost can be amortized across many predictions on the same GPU.

2.4 Training data and augmentation

Training data: full PWM-LDCT (to be confirmed, 2027a) training split with paired (full-dose, low-dose) reconstructions at $r \in \{0.10, 0.25, 0.50\}$. We mix all three dose levels in each minibatch with weights $(0.4, 0.4, 0.2)$, biased toward the most clinically relevant levels.

Augmentation: random flip ($p = 0.5$), random rotation in $[-5^\circ, 5^\circ]$, random window-level shift in $[-50, 50]$ HU. No augmentation that violates physical consistency (no random noise injection on top of the already-noisy low-dose, no patch shuffling).

3 Experiments

Datasets. Training: PWM-LDCT (to be confirmed, 2027a) training split ([TODO: N] patients, [TODO: N] scans). Evaluation: PWM-LDCT test split ([TODO: N] patients), AAPM-2016 (AAPM, 2016) held-out scans ([TODO: N] patients), and LIDC-IDRI (Armato III et al., 2011) sub-sample ([TODO: N] patients) for downstream task validation.

Baselines. Three baseline reconstruction methods, reproduced within ≤ 0.5 dB PSNR of their published numbers:

- RED-CNN (Chen et al., 2017): foundational encoder-decoder CNN, 2017.
- A transformer-based reconstruction ([TODO: select: CTformer / TransCT / Restormer-LDCT]).
- A diffusion / score-based reconstruction ([TODO: select: 2024 score-LDCT]).

Cross-vendor splits. We construct three leave-one-vendor-out splits: train-on-{GE, Canon}/test-on-Siemens; train-on-{Siemens, Canon}/test-on-GE; train-on-{Siemens, GE}/test-on-Canon. Each cross-vendor experiment uses the full training set with the held-out vendor’s patients removed (patient-level, not slice-level).

Metrics. (i) PSNR, SSIM, LPIPS (Zhang et al., 2018) on the test reconstructions; (ii) lung-nodule detection AUC at fixed FPR = 0.1, using a pre-trained nodule detector frozen across reconstruction methods; (iii) per-pixel uncertainty calibration — Spearman correlation of $\sigma(y)_{ij}$ against $|\hat{x}(y)_{ij} - x_i^{\text{ref}}|_{ij}$ on held-out test; (iv) inference latency (seconds per slice on a single NVIDIA A100), peak GPU memory; (v) 5-tuple signal-equivalence credentials per the framework of to be confirmed (2027b) at $r \in \{0.10, 0.25\}$ for lung-nodule detection, $\varepsilon = 0.02$, $\alpha = 0.05$, subpopulation $\Pi = \text{adult-chest test split}$.

4 Results

Reconstruction quality. Table 2 [TODO: populate] reports PSNR, SSIM, and LPIPS for each method at each dose level on the PWM-LDCT test split. The reference method achieves PSNR within [TODO: X] dB of the diffusion-based baseline at $r = 0.25$, while consuming [TODO: Y]× less inference time and using [TODO: Z]× less GPU memory.

Method	$r = 0.10$		$r = 0.25$		$r = 0.50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
FBP	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RED-CNN	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Transformer	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Diffusion	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
PWM ref	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 2: Reconstruction quality on the PWM-LDCT test split. Values to fill at D9+270 after WS-3 Phase 3 (v1 deep ensemble) lands.

Cross-vendor generalization. Table 3 reports the PSNR drop Δ when each method is trained without one vendor and evaluated on that held-out vendor’s test scans. The reference method’s Δ should be ≤ 2 dB; baselines are not designed for cross-vendor generalization and we expect $\Delta \approx 3\text{--}5$ dB.

[TODO: TABLE: Cross-vendor PSNR drop (Δ , dB) under leave-one-vendor-out training; lower is better]

Table 3: Cross-vendor PSNR drop (Δ , dB) under leave-one-vendor-out training; lower is better (placeholder; populate when data lands).

Uncertainty calibration. The deep ensemble’s per-pixel standard deviation correlates with per-pixel error at Spearman $\rho =$ [TODO: populate]. A reliability diagram is provided as Figure 1.

[TODO: FIGURE: Reliability diagram: predicted per-pixel standard deviation (ensemble) vs observed per-pixel reconstruction error on held-out test]

Figure 1: Reliability diagram: predicted per-pixel standard deviation (ensemble) vs observed per-pixel reconstruction error on held-out test (placeholder; populate when data lands).

Downstream task performance. Lung-nodule detection AUC against a frozen detector is reported in Table 4. The downstream-task results may diverge from pixel-wise results: a method with higher PSNR may produce lower task AUC if it over-smooths small lesions (Kim et al., 2019). Showing both is the point.

[TODO: TABLE: Lung-nodule detection AUC at fixed FPR = 0.1, per method, per dose level, using a frozen pre-trained detector]

Table 4: Lung-nodule detection AUC at fixed FPR = 0.1, per method, per dose level, using a frozen pre-trained detector (placeholder; populate when data lands).

4.1 Signal-equivalence credentials

Per the framework of to be confirmed (2027b), we report 5-tuple credentials for the reference method:

These credentials are the canonical artifact that future methods will report against. Reporting only PSNR/SSIM/LPIPS is insufficient; reporting only verdicts loses the CI; reporting both is the standard we propose.

5 Conclusion

We have presented an open-source reference reconstruction method for low-dose CT, designed not to maximize PSNR but to be *credible, improvable, and permissively licensed*: cross-vendor evaluated, deep-ensemble UQ-calibrated, signal-equivalence-credentialed per (to be confirmed, 2027b), and Apache-2.0 licensed. The method is the seed entry of the permanent PWM Low-Dose CT Challenge leaderboard (to be confirmed, 2028); an L4 cert anchoring the RunBundle CID is registered on the PWM blockchain mainnet. We invite the community to compete by improving any of the four independent axes (denoiser, forward model, iteration schedule, UQ method) without re-deriving the full stack. Code, weights, and RunBundle are released under Apache 2.0 at https://github.com/integritynoble/low_dose_CT/tree/main/WS-3_reference_method.

Acknowledgments

[TODO: Fill at submission.]

r	Task	ε	α	Subpopulation	Verdict	CI
0.25	Lung nodule AUC	0.02	0.05	adult-chest test split	[TODO:]	[TODO:]
0.10	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]
0.25	Liver lesion Dice	0.03	0.05	adult-abdomen test split	[TODO:]	[TODO:]

Table 5: Signal-equivalence credentials computed per [to be confirmed](#) (2027b) against full-dose FBP as the reference. Verdicts: PASS / FAIL / INDETERMINATE.

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